ONE-PAGE BRIEF

# Post-quantum cryptography review: monero @ 3fd0f88c20f5

Client: Monero — public repository (passive review) · Source: github.com/monero-project/monero, commit `3fd0f88c20f5` · Reviewed: 2026-10-28 16:45 UTC · Risk profile: Crypto exchange, custodian or wallet

POST-QUANTUM MATURITY

**Level 0 of 4**

no post-quantum cryptography

FINDINGS

**0** Critical **0** High **5** Medium **2** Low **1** Info

QUANTUM TIMELINE (MOSCA: X + Y − Z)

Key establishment: 10 + 5 − 8 = 7 → **too late**

Signatures: 10 + 5 − 8 = 7 → **too late**

**Next step:** Inventory where classical public-key cryptography protects long-lived data and keys, and choose the first migration: usually a hybrid TLS key exchange and wrapped keys.

## WHAT MATTERS MOST

| | | |
|---|---|---|
| Medium | QSI-2026-05417 | **Transaction signing for 1 chain(s) relies on elliptic-curve signatures**<br>What to do: Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains... |
| Medium | QSI-2026-05414 | **Quantum-vulnerable Ed25519: role not determined (4 file(s))** preliminary<br>What to do: Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form... |
| Medium | QSI-2026-05415 | **Quantum-vulnerable X25519: role not determined (2 file(s))** preliminary<br>What to do: Classify each usage, record data lifetimes, and plan migration to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in... |
| Medium | QSI-2026-05416 | **Quantum-vulnerable ECDH: role not determined (2 file(s))** preliminary<br>What to do: Classify each usage, record data lifetimes, and plan migration to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in... |

3 more in the full report.

## WHAT THE QUANTUM-VULNERABLE CRYPTOGRAPHY PROTECTS

| Role | Algorithms | What to do |
|---|---|---|
| Medium transaction and message signing | Ed25519 | Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum... |
| Medium role not determined | Ed25519, ECDH, X25519 | Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205)... |

## KNOWN VULNERABLE COMPONENTS

Advisories for the versions in use: 3; unlikely to apply to this build: 1 (a Cargo feature that is off, affected functions not named in the code, or a platform that is not built); to review: 2.

Static review of source code and manifests; files analysed: 580. No requests were sent to the owner's systems.

Full report, content SHA-256: `c640f43c1f7c8aeef4a0c7208351c7fb5f0d46ab7c3a0401b7c2139ece61dd32`

Full report signature (report.sig.json): Ed25519 key, SHA-256 fingerprint `5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`. bundle.sig.json lists the SHA-256 of every file in this bundle and is signed too; verify-bundle checks it offline.

Preliminary findings need confirmation in the code; the full report gives the evidence, the limits of each check and how to re-test.